

Task 10 - File Input Output

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COS30031 Games Programming

Part A:

Notes:

- Created Data header file.
- Created an instance and set its values in main.
- Created a reusable print function in main.
- Modified main to create and save files.
- Implemented read from file function.
- Organized Main Code Structure.
- Committed and pushed each implementation.
- Answered the 3 questions.

Questions:

1. There are different file open modes: What are they?

The Different files open modes would be **"ios::in"** to read **"ios::out"** to write, **"ios::binary"** to use binary operations, **"ios::ate"** to end at opening, **"ios::app"** to append data to file and **"ios::trunc"** to replace and delete previous content in file when opening.

2. What happens if you don't "close" the file? Is it something we need to worry about?

when a file does not get close can lead to data corruption.

3. How many bytes are in the file? Is this expected based on the size of the variable types?

the file has 12 bytes which means if 1 byte is for a char and int and float typically have 4 bytes each which should make up 9 bytes the discrepancy is due to padding where sometimes c++ compilers can add padding to ensure that data structures align to memory for performance reasons and to remove padding we can add **"#pragma pack"**.

Part B:

Notes:

- Commented Out Part A Code
- Undid previous comments and made a new solution for Part B
- Created test.txt file.
- Was able to Open up text file.
- Modified code to ignore blank lines, ignore line that starts with #, split other lines and print each split line to screen.

Part C:

Notes:

- Created New Solution for Part C
- Added json.hpp library.
- Print JSON result on console.